

Joel Biswas

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EDUCATION

Georgia Institute of Technology

Bachelor of Science in Electrical Engineering; Senior Classification; GPA: 4.00/4.00

Atlanta, GA

Exp. May 2028

EXPERIENCE

Digital Design and Verification Member

September 2026 – Present

Georgia Tech SiliconJackets

Atlanta, GA

- Member of Georgia Tech's student ASIC design team developing custom integrated circuits through RTL design, verification, and physical design.
- Completing a 32-bit RISC-V processor design in SystemVerilog as preparation for contributing to the team's annual chip.
- Developing a SystemVerilog verification environment with constrained-random instruction generation, assertions, monitors, and architectural state checking to validate processor functionality and corner cases.

Electrical Engineering Intern

June 2025 – June 2026

Origami Robotics LLC

Suwanee, GA

- Designed a custom 4-layer ESP32-C3 PCB integrating CAN, dual I²C GPIO expanders, USB-C, onboard power regulation, and 30 GPIO channels.
- Programmed and tested embedded hardware interfaces, validating peripheral communication, power delivery, and controller functionality during system bring-up.
- Designed power, signal-integrity, and protection circuitry including 3.3 V buck regulation, IC decoupling, USB-C TVS protection, and differential routing for USB and CAN.

Team Captain & Electrical Lead

May 2025 – May 2026

Lambert Robotics FRC Team 9477

Suwanee, GA

- Led a ~60-member engineering team across electrical, programming, CAD, manufacturing, and system integration in the development of competition robots.
- Designed and debugged robot-wide electrical systems integrating CAN networks, sensors, motor controllers, power distribution, and embedded control hardware.

PROJECTS

32-bit RISC-V Processor | *SystemVerilog, Cadence Xcelium*

September 2026 – Present

- Designed a 32-bit RISC-V processor in SystemVerilog, implementing instruction decode, register file, ALU, memory interfaces, and control logic for arithmetic, shift, load/store, and branch instructions.
- Developed and debugged RTL through simulation and testbenches in Cadence Xcelium, validating instruction execution and memory transactions against expected architectural state.

Custom R2-D2 Robot | *ESP32, C++, nRF24L01+, CAD*

July 2025

- Designed and built a mobile three-wheel robotic platform integrating DC gearmotors, motor drivers, servos, power regulation, and stepper-driven dome rotation.
- Developed wireless embedded control using microcontrollers and nRF24L01+ transceivers, implementing joystick-based drive control and remote actuation.
- Iteratively redesigned the drivetrain after identifying insufficient motor torque in the initial prototype, recalculating mechanical requirements and selecting higher-torque geared motors to achieve reliable locomotion.

TECHNICAL SKILLS

HDL/Programming: SystemVerilog, Verilog, C++, Python, C

Digital Design: RTL Design, RISC-V, Digital Logic, Computer Architecture, FSMs, Static Timing Analysis

Embedded Systems/Hardware: PCB Design, Embedded Systems, ESP32, CAN, SPI, I²C, UART, PWM, Circuit Prototyping

Tools: Cadence Xcelium, KiCad, Git, Linux, VS Code, Fusion 360, Onshape